S.P.M College,Udantpuri

Bachelor Of Computer Application (BCA)

Part -1 (Paper-1) – **Hira Kumar**

COMPUTER ORGANIZATION

❖ What is Computer Organization ?

- Computer Organization is the study of the internal structure, components, and working of a computer system. It explains how different hardware units—such as the CPU, memory, input/output devices, and control unit—are arranged, connected, and work together to execute instructions and process data. i.e. Learn about Functional units and Working flow.
- It is also called Block digram of Computer / Element of Computer / Components of Computer / How to work a Computer

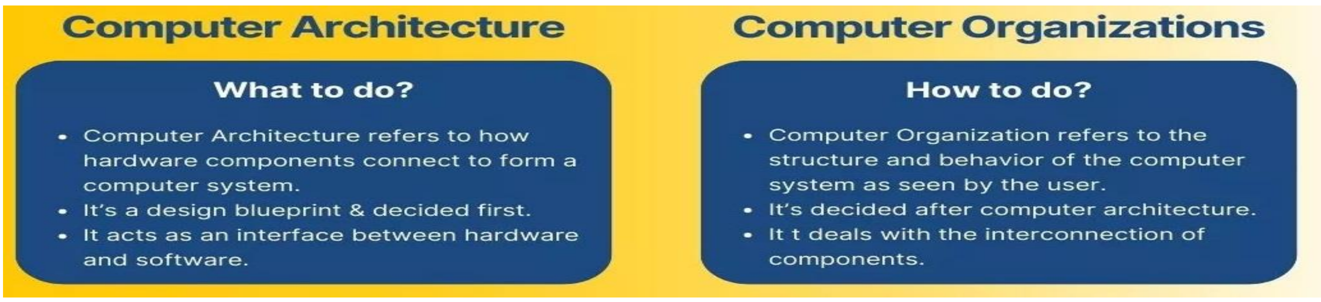
❖ What is Computer Architecture ?

Computer Architecture is the study of the design and structure of a computer system as visible to the programmer. It defines what the computer can do, rather than how it is physically implemented.

✓ **Computer Organization** → How it Works (कंप्यूटर कैसे काम करता हैं), Logical Design

✓ **Computer Architecture** → What it can do (कंप्यूटर क्या काम कर सकता हैं), Physical Implementation

- **Human Body Architecutre → for see – Eye, for think– mind, for hear-Ear, For work -hand, for wove -leg,…….**
- **Computer Architecture → Instruction Set, Data type, 32bit/46bit, Addressing mode**
- **Human Body Organization → Internal working, Heart कैसे pump करेगा, nerves system कैसे भेजती हैं, खाना कैसे digest होगा...**
- **Computer organization → ALU कैसे काम करता हैं,cache,RAM से कैसे जुडी है,.....**



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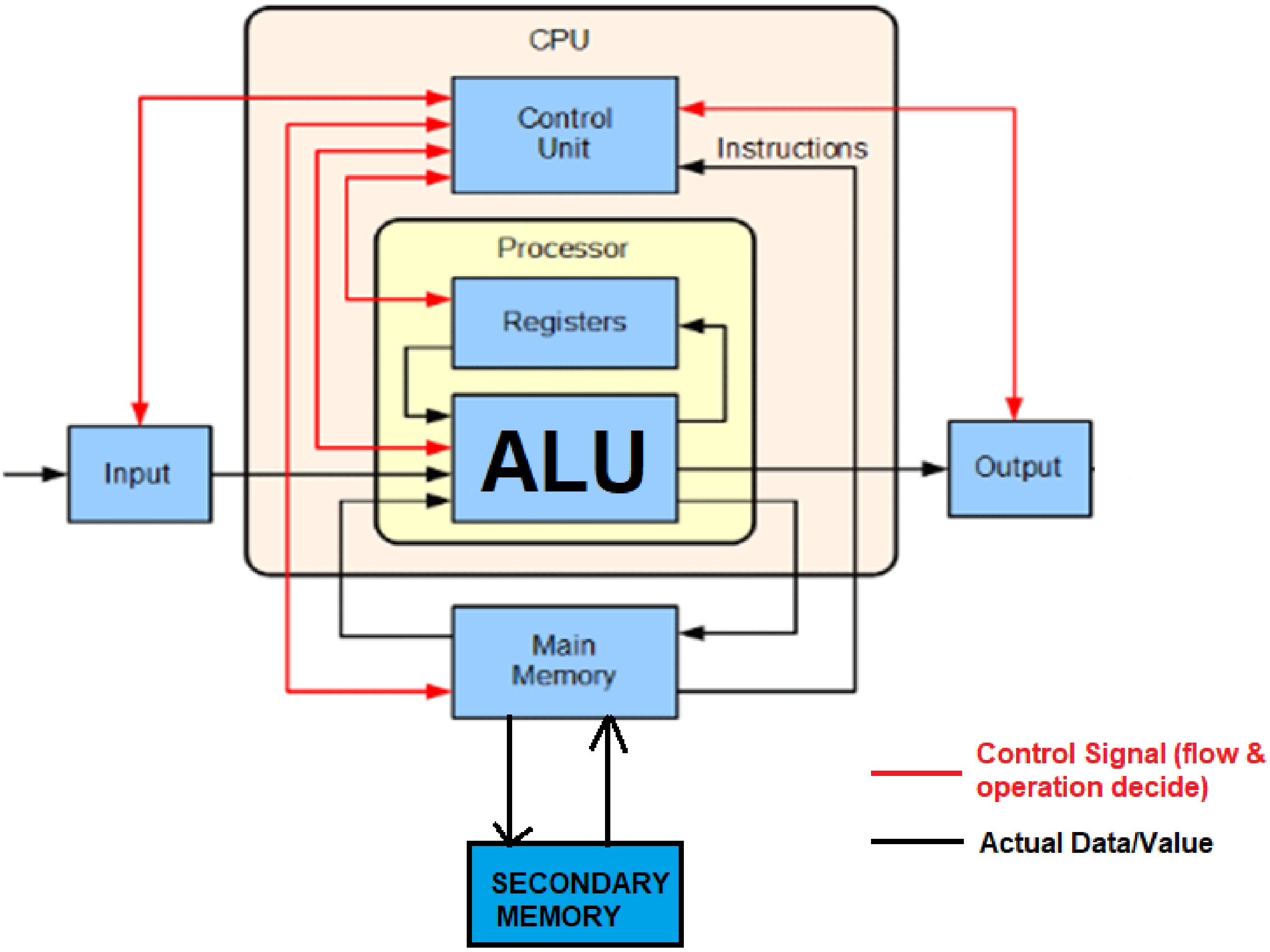
❖ Basic units of Computer

Total Four to Five (depending on how you count) units of Computer.

i. Input Unit
ii. Output Unit
iii. Processing Unit (CPU)
    a) ALU (Arithmetical & Logical Unit)
    b) CU (Control Unit)
iv. Memory Unit (Both – Main + Secondary)

Note :- if 5th then (i) Input (ii) Output (iii) ALU (iv) CU (v) Memory

Block Diagram of Computer OR Basic Organizaation of Computer OR Logical Diagram of Computer OR Functional Organization of Computer



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1. Input Unit

- The devices through which data and instructions are entered into the computer are called input.
- Example – Keyboard, Mouse,Joystick, Light pen, Scanner, Microphone, Webcam, Magnetic Ink Character (MICR), OMR (Optical Mark Reader), BCR (Bar Code Reader), Touch Screen, Smart Card Reader, Bio-metric Sensor...etc

2. Output Unit

- The device through which the processed results (i.e. Information) by the computer can be seen or heard (देखना या सुनना) is called output device.
- e.g. – Monitor, Printer,Speaker, Plotter, Projector, Head Phones,...

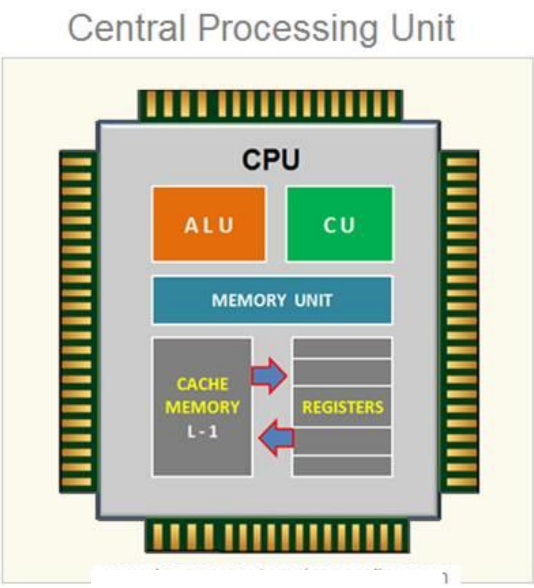
3. Processing unit

CPU is that part of the computer in which Arithmetical, logical and Control input/output operations and instructions are decoded and than executed.

Or

The Unit where all the processing is done is called as Central Processing Unit.

- ✓ CPU is called the brain of computer.
- ✓ The CPU is build as a single Integrated Circuit (IC) and is also known as microprocessor.
- ✓ CPU of micro computer is also called micro-processor.
- ✓ CPU also Called – Processor OR MicroProcessor
- ✓ Speed of CPU is Measured in
  - Hertz
  - Gigahertz (GHz)
  - Megahertz (MHz)
  - MIPS (Million Instruction per second)
  - KIPS (kilo instruction per second)
- ✓ CPU Understand which language – Machine OR Binary
- ✓ CPU Speed of PC is – 100 KIPS
- ✓ First microprocessor Intel 4004, In which all component of CPU available in a single Chip.
- ✓ E.g. – Intel 4004, Intel duo core, Pentium IV etc



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Two Main components of a CPU are the –

1. Arithmetic Logic Unit (ALU)
2. Control Unit (CU)

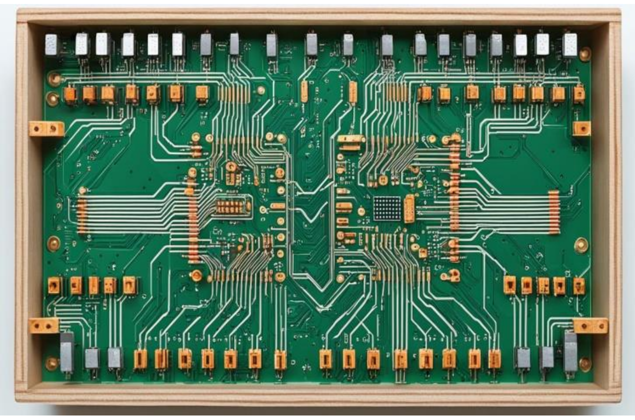
Other Component is –

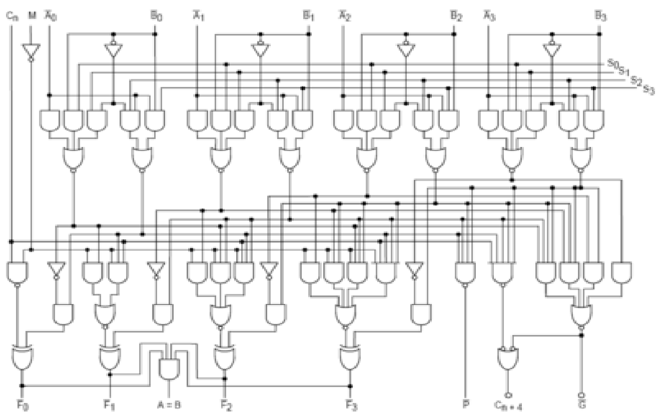
i. Register
ii. Cache memory
iii. Million Transistors

All part available/printed on Silicon die

➢ ALU (Arithmetics and Logical Unit)

- The ALU is the place of CPU where actual data & instruction are processed / Execute.
- Performs all arithmetical & logical operations.
- An arithmetic operation contains basic Calculating operations like addition, subtraction, multiplication, division.
- Logical operations contains comparison such as less than, greater than, less than equal to, greater than equal to, equal to, not equal to, AND, NOT, OR, XOR.
    - Arithmetic operations: + − × ÷
    - Logical operations: AND, OR, NOT, XOR
    - Comparison operations: <, >, ≤, ≥, =
- **ALU uses registers to hold the data that is being processed.**
- **Very Fastest Memory – Registers.**





➢ CU (Control Unit)

- Control unit control the movement of data and instructions into and out of the CPU, and to control the operations of the ALU.
- In sort, its main function is to manage all the activities in the computer system.
- Control unit control the internal parts as well as the external parts related with the computer.

Parts/Unit of CPU = 3

1. **CU = Control Unit**
2. **ALU = Arithmetic Logical Unit**
3. **MU = Memory Unit**

4. Storage Unit / Memory Unit

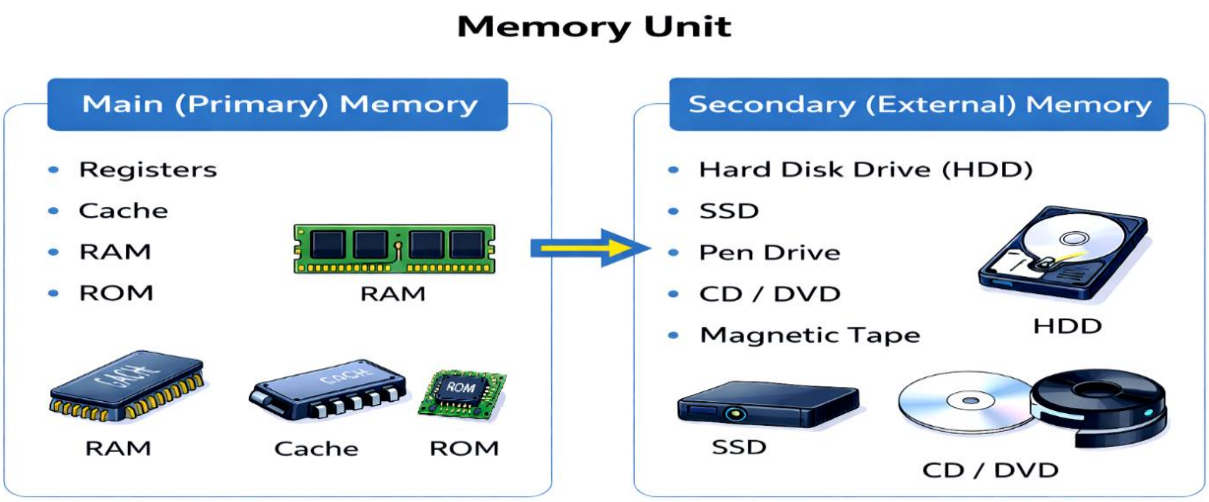
- Memory Unit is that part of the computer in which data & instruction have to store inside the computer, before the actual processing start. Same way the result of the computer must be stored before passed to the output devices.
- The **Memory Unit** stores:

✓ Program instructions

✓ Input data

✓ Intermediate results

✓ Final output before transfer to output devices



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